THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student finds that a colourless liquid turns blue litmus red and does not affect red litmus. It also fizzes with baking soda. The liquid is: 1

(a) a base **(b)** neutral **(c)** a salt **(d)** an acid

2. In an experiment, 20 mL of a base is neutralised by 20 mL of a certain acid of the same type. If 40 mL of the same base is to be neutralised, the acid needed is: 1

(a) 10 mL **(b)** 40 mL **(c)** 20 mL **(d)** 80 mL

3. A rainwater sample turns blue litmus red. The rain may be: 1

(a) acidic **(b)** salty **(c)** basic **(d)** neutral

4. A liquid turns turmeric red. When liquid Y is added, the colour changes back to yellow. Liquid Y is: 1

(a) a base **(b)** sugar **(c)** water **(d)** an acid

5. In a neutralisation, if 15 mL of acid is needed for 25 mL of a base, then for 50 mL of the base it will be: 1

(a) 25 mL **(b)** 30 mL **(c)** 60 mL **(d)** 15 mL

6. Which statement about milk of magnesia is correct? 1

(a) it is an acid used to relieve acidity **(b)** it is a neutral salt used as a drink
(c) it is an indicator **(d)** it is a base used to relieve acidity

7. Liquid A turns China rose pink, liquid B green. If A and B are mixed in equal strengths, the mixture most probably: 1

(a) turns almost neutral **(b)** becomes more acidic
(c) turns blue **(d)** becomes more basic

8. A farmer needs 200 kg of lime for 1 hectare of acidic soil. For 2.5 hectares, the lime needed is: 1

(a) 400 kg **(b)** 500 kg **(c)** 250 kg **(d)** 800 kg

9. **Assertion (A):** Turmeric paper turns red when soap solution is dropped on it. 1
Reason (R): Soap solution is basic in nature.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Red litmus paper turns blue when dipped in lemon juice. 1
Reason (R): Lemon juice is acidic in nature.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. When a few drops of lemon are added to a basic solution containing turmeric (red), the colour becomes yellow. Explain. 2

12. How can you distinguish between two colourless liquids, one acid and one base, with a single indicator? 2

13. Explain why a neutral solution does not change the colour of litmus. 2

14. Why does the stomach not get damaged by its own hydrochloric acid? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Student X finds that 10 mL of acid is neutralised by 20 mL of base. How much acid neutralises 60 mL of base of the same strength? 3

16. Explain why we use different indicators for different solutions. 3

17. Write the neutralisation equation for the formation of potassium sulphate. 3

OR

A soil test shows it is acidic. List three ways to treat it and explain. 3

18. Explain why a student's mixture of acid and base became warm. 3

OR

Explain how red cabbage juice could be used to compare the strength of lemon and vinegar. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** After eating too many spicy samosas at a mela in Lucknow, Rohan complains of burning in his stomach. His mother gives him a spoonful of milk of magnesia. Rohan's younger sister suggests giving him lemon juice instead, saying it is 'good for the stomach'. 5
(i) Name the acid naturally present in the stomach.
(ii) What is the nature of milk of magnesia?
(iii) Name the type of reaction that gives relief to Rohan.
(iv) Explain why lemon juice would not be a good choice.

20. Explain, with the help of examples, how the ideas of acids, bases and neutralisation connect agriculture, health and the environment. 5

OR

Evaluate whether natural indicators or lab indicators are better for use at school. 5

END OF QUESTION PAPER — Total: 40 Marks